



School of Computer Science, Engineering and Applications (SCSEA)

B. Tech TY (CCSA)

Subject: Blockchain Systems

Name of the Student: Shweta shelke

PRN: 20220802278

Title Of Practical: Create a simple proof-of-work blockchain in a programming language of choice. Demonstrate the mining process and block validation.

Introduction

A blockchain is a distributed ledger that consists of a chain of blocks, each containing transaction data, a timestamp, and a cryptographic hash. Proof-of-Work (PoW) is a consensus algorithm that requires miners to solve a computational puzzle to validate transactions and add new blocks to the chain. This process ensures security and prevents malicious activities such as double-spending.

Solidity Program

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract PoWBlockchain {
5     struct Block {
6         uint index;
7         bytes32 previousHash;
8         bytes32 transactions;
9         uint nonce;
10        bytes32 hash;
11    }
12
13    Block[] public blockchain;
14    uint public difficulty = 2; // Number of leading zeros required in hash
15
16    event BlockMined(uint index, bytes32 hash);
17
18    constructor() {
19        _createGenesisBlock();
20    }
21
22    function _createGenesisBlock() private {
23        bytes32 genesisHash = keccak256(abi.encodePacked(block.timestamp, "Genesis Block", uint(0)));
24        blockchain.push(Block(0, 0x0, "Genesis Block", 0, genesisHash));
25    }
26}
```

```
contract PoWBlockchain {
    struct Block {
```

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```

    uint index;
bytes32 previousHash;
bytes32 transactions;
    uint nonce;
    bytes32 hash;
}

Block[] public blockchain;
uint public difficulty = 2; // Number of leading zeros required in hash

event BlockMined(uint index, bytes32 hash);

constructor() {
    _createGenesisBlock();
}

function _createGenesisBlock() private {    bytes32 genesisHash =
keccak256(abi.encodePacked(block.timestamp, "Genesis Block", uint(0)));    blockchain.push(Block(0,
0x0, "Genesis Block", 0, genesisHash));
}

function mineBlock(bytes32 transactions) public {
    Block memory lastBlock = blockchain[blockchain.length - 1];    uint
nonce = 0;
    bytes32 newHash;

    while (true) {        newHash = keccak256(abi.encodePacked(lastBlock.hash,
transactions, nonce));        if (_isValidHash(newHash)) {
break;
        }
        nonce++;
    }

    blockchain.push(Block(blockchain.length, lastBlock.hash, transactions, nonce, newHash));
emit BlockMined(blockchain.length - 1, newHash);
}

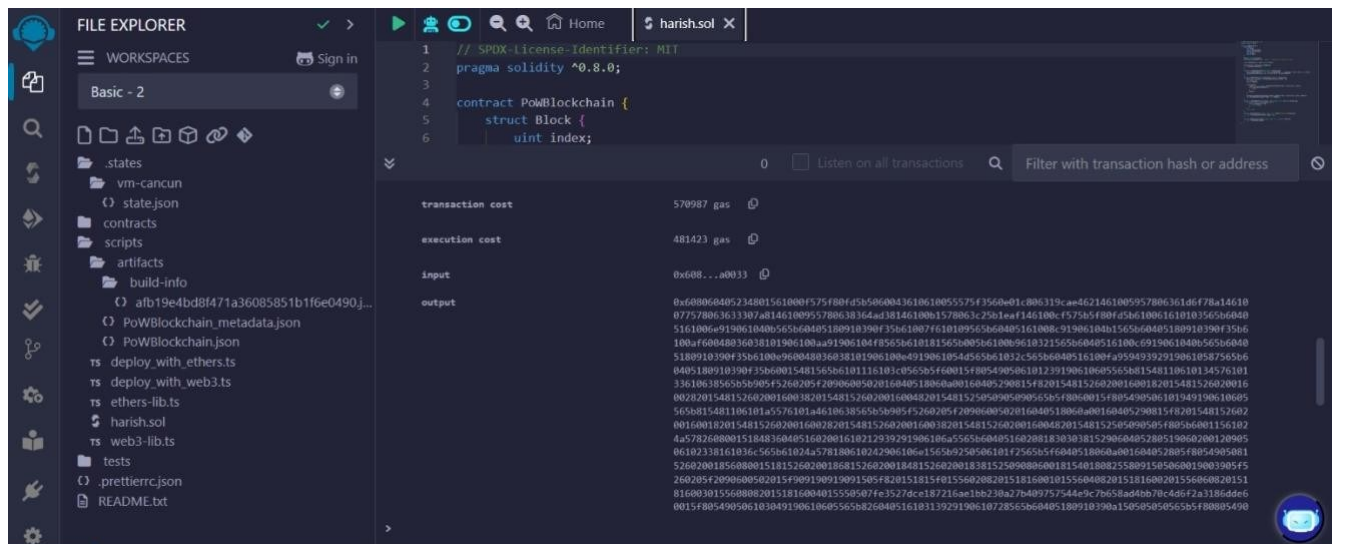
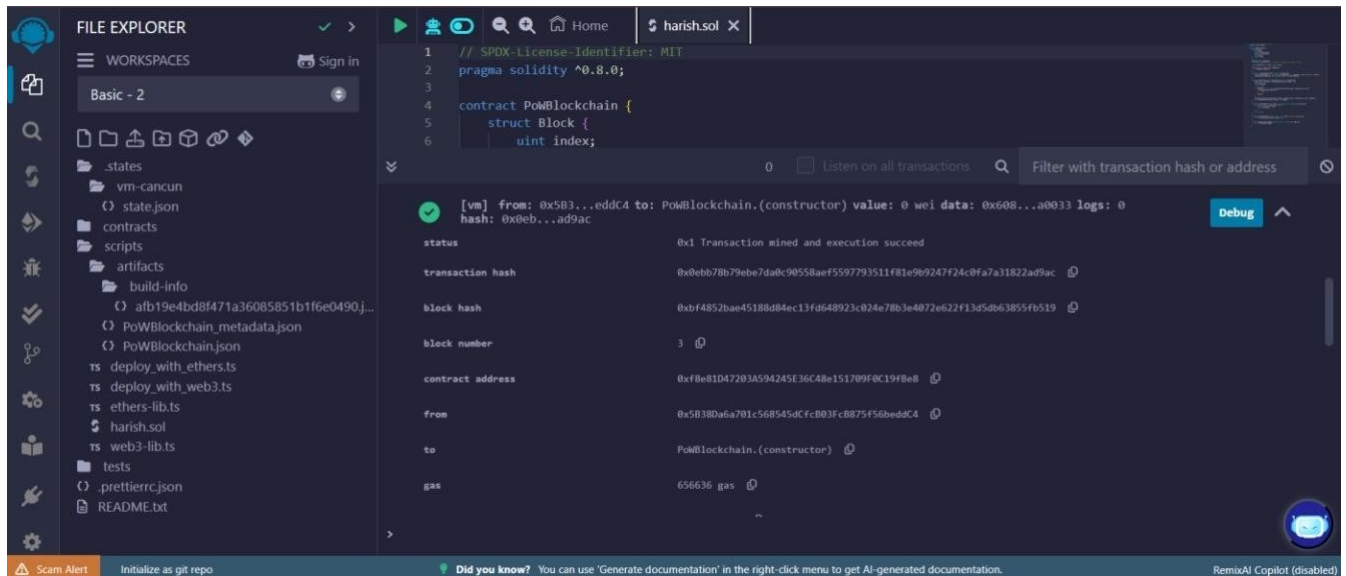
function _isValidHash(bytes32 hash) private view returns (bool) {
for (uint i = 0; i < difficulty; i++) {        if (uint8(hash[i]) != 0) {
return false;
        }
    }
    return true;
}

function getLatestBlock() public view returns (Block memory) {
return blockchain[blockchain.length - 1];
}

function getBlockchainLength() public view returns (uint) {
return blockchain.length;
}
}

```

Program Output



- **Status: 0x1** ○ The transaction was mined and executed successfully.
- **Transaction Hash: 0x0ebb78b7...22ad9ac**
 - A unique identifier for this transaction.
- **Block Hash: 0xbfa852b...55fb519** ○ The hash of the block containing this transaction.
- **Block Number: 3** ○ This transaction was included in block #3.
- **Contract Address: 0xf8e81D...19fBe8** ○ The address of the deployed smart contract.

- **From: 0x5B38Da6...6beddC4** ○ The sender (account that deployed the contract).
- **To: PoWBlockchain.(constructor)** ○ Indicates contract deployment (not a transaction to an existing address).
- **Gas Used: 656636** ○ The total gas consumed by this transaction.
- **Transaction Cost: 570987 gas** ○ The actual gas spent to execute the transaction.
- **Execution Cost: 481423 gas** ○ The gas required for executing contract code.
- **Input Data: (0x608...a0033)** ○ The bytecode of the smart contract deployed.
- **Output (0x6080604052...0033)** ○ The compiled bytecode of the deployed contract.

Conclusion

In this lab, we implemented a simple blockchain using the Proof-of-Work consensus mechanism. We demonstrated how mining works by solving a computational puzzle and how blocks are validated using cryptographic hashes. The lab provides a foundational understanding of blockchain principles and how PoW ensures security and integrity in decentralized networks.